

Zerodha: Improving the retail investment experience with seamless collaboration on Google Workspace

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT ZERODHA



About Zerodha

Zerodha enables employees to work remotely during the 2020 lockdown with Google Workspace, enabling seamless access to enterprise tools while enabling re

Founded in 2010, Zerodha is one of India's largest stock brokers, serving millions of users. The company offers investors attractive brokerage rates for all types of investments, including stocks, mutual funds, and commodities. The name Zerodha borrows from the English word, Zero, and Rodha, which means barrier in Sanskrit. Zerodha removes barriers for traders by charging a flat fee of 20 Indian Rupee, the equivalent of 27 cents, rather than a percentage commission in investment size. The company donates a portion of its profits to the

meetings anywhere Google Meet.

Rainmatter Foundation, an NGO dedicated to combating climate change.

Industries: Financial Services & Insurance

Location: India

Results

- Collaborating on Docs for quicker development of blogs and company newsletters
- Transitioned 1,000+ staff with minimal effort to full remote work overnight using Google SSO with 2FA
- Onboards millions of users in 4 weeks
- Automate document verification with Vision AI to speed up application process

Enabl
acco
openi
minut
of day

, Gmail

(<https://workspace.google.com/products/gmail/>)

Google Workspace

(<https://workspace.google.com/>)

(<https://workspace.google.com/products/chat/>)

Docs

(<https://workspace.google.com/products/docs/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

Google Chat

(<https://workspace.google.com/products/chat/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Google Cloud (<https://cloud.google.com/>)

Vision AI (<https://cloud.google.com/vision/>)

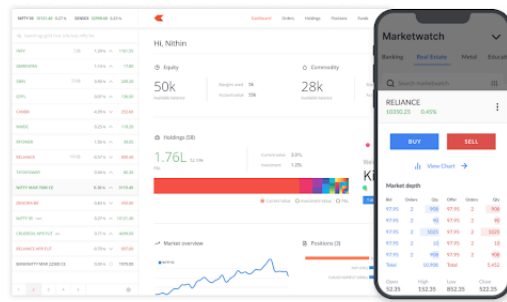
The rise of smartphone ownership in India in recent years alongside the pandemic has brought about a surge in retail investing. More people are choosing to enter the stock market through their mobile devices due to the ease of access. In fact, the Indian National Stock Exchange (NSE) reported that mobile trading jumped from less than 7% in 2019 to more than 19% in 2021. Zerodha (<https://zerodha.com/>), one of the largest online brokerages in India, experienced a double-digit growth in customers every month since the outbreak in 2019.

Featuring low brokerage fees and easy-to-use apps, Zerodha simplifies trading for anyone with a smartphone. By reducing manual tasks, the company keeps margins low and invests in developing digital solutions such as its flagship trading platform, Kite, and its mutual fund investing platform, Coin.

When Zerodha began operations in 2010, the team was eager to eliminate cumbersome tasks related to legacy solutions, such as managing mail servers. According to Vishnu Sudhakaran, full-stack developer at Zerodha, Google Workspace (<https://workspace.google.com/>) was the obvious choice. This was because Google Workspace was easy to set up for employees as they were already familiar with the Google ecosystem and did not require any training.

“Zerodha's guiding principle is scalability. Our trading platform grew to millions of users with the help of a

small tech team,” says Vinay Kumar, AVP Technology at Zerodha. “Rather than managing email accounts and servers, we want engineers to focus on building digital solutions that our customers need.”



Team collaboration in the new normal with Google Workspace

Google Meet

(https://workspace.google.com/intl/en_sg/products/meet/)

is now an integral part of life at Zerodha to help colleagues stay connected. Virtual meetings are a fairly new practice at Zerodha, as it only started to become a norm since the COVID-19 pandemic. Before, everyone would go into the office to work, so meetings were often face-to-face.

Employees use Meet for daily project meetings, monthly catch-ups, and even use them for social events, such as playing multiplayer games with team members. With the seamless integration of Meet with Google Chat

(https://workspace.google.com/intl/en_sg/products/chat/)

, employees can also choose to start a meeting

instantly through the Chat platform. This allows impromptu meetings to take place, whether it is a one-on-one or group meeting.

One of the value propositions that Zerodha provides is to empower traders through financial literacy. The marketing team does this by creating blog posts around topics like business valuation and option trading, that aim to educate customers, as well as generate new leads.

Using Docs

(https://workspace.google.com/intl/en_sg/products/docs/)

, employees can view, edit, and share articles that are a work in progress internally in real time, so that they can get immediate feedback and input from each other all in the same document. They also use Docs to share meeting notes where necessary.

“Zerodha's
guiding principle
is scalability. Our
trading platform
grew to millions
of users with the
help of a small
tech team.
Rather than

managing email
accounts and
servers, we want
engineers to
focus on
building digital
solutions that
our customers
need.”

—Vinay Kumar, AVP
Technology, Zerodha

Improving security and ease of access management with Google SSO

Protecting company and customer data is a top priority for Zerodha. For data protection beyond passwords, the company implemented Google Single Sign-On

(<https://cloud.google.com/architecture/identity/single-sign-on>)

(SSO) with Two-Factor Authentication (2FA) in compliance with Securities and Exchange Board of India (SEBI) cybersecurity guidelines.

Employees now access enterprise apps with a single login using Google SSO and 2FA. To reduce security risks, employees who leave the company would have their Google account suspended, which in turn

automatically disables access to all enterprise applications.

"Employees have access to different systems depending on their roles. SSO with 2FA from Google simplifies user authentication and access from any location and device," says Sudhakaran. "Due to this architecture, we could move to work from home practically overnight with minimal impact on customers." Even though pandemic restrictions are easing, Zerodha continues to support work from home as the default working arrangement.

As of November 2021, Zerodha upgraded its 1,000+ employees to the Enterprise plan to enhance security with features such as data loss protection (DLP), which prevents users from sharing sensitive data, and alerts administrators when users log in from unfamiliar devices.

"Employees have
access to
different
systems
depending on
their roles. SSO
with 2FA from
Google

simplifies user authentication and access from any location and device. Due to this architecture, we could move to work from home practically overnight with minimal impact on customers.”

—Vishnu Sudhakaran, Full Stack Developer, Zerodha

Scaling up digital onboarding to meet regulatory changes

Each day, Zerodha receives around 25,000 account opening applications, all of which undergo a Know-Your-Customer (KYC) process as required by SEBI in order to minimize fraudulent activities such as identity theft and money laundering. Customers must provide KYC documents such as their Permanent Account Numbers (PAN) cards, to verify their identities. Pre-pandemic, there were about 2,500 accounts opened in a day, currently, we see around 15,000 accounts opened per day.

To simplify the KYC process, Zerodha deployed Vision AI (<https://cloud.google.com/vision>) in April 2020, which analyzes scanned copies of PAN cards submitted by users as proof of identity. Using optical character recognition, the system detects text in the images, extracts the PAN names and numbers and compares them to the information the clients themselves provided in their application forms. If their PAN cards don't match the information they provided, users receive an alert and will need to submit the correct documentation in order to open their accounts.

"Google Vision API helps us to validate the PAN card uploaded by the user in real-time, thereby avoiding a manual intervention from the support agent and this helps to significantly reduce the time spent on account opening for new users. By eliminating 80% of the manual process, we continue to improve the customer experience as we grow."

